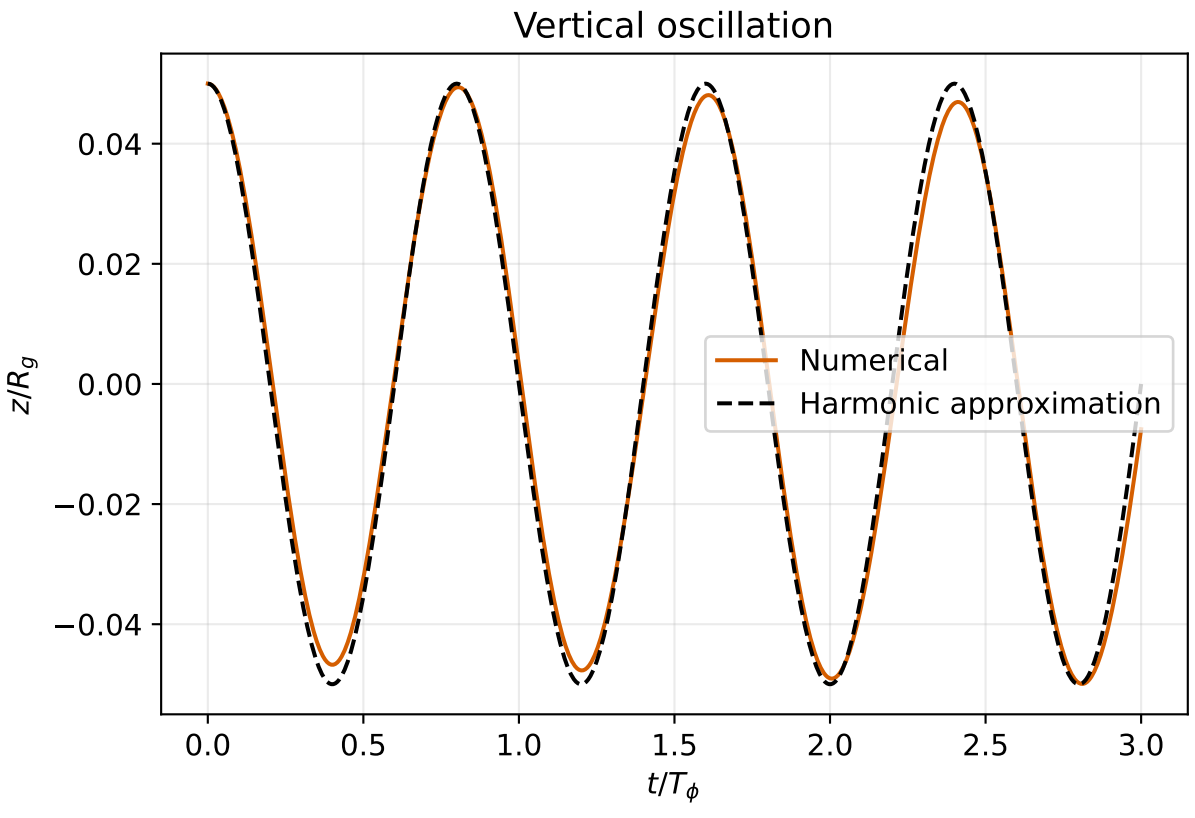
Vertical oscillation
0.04
0.02
0.00
−0.02
−0.04
z/R_g
0.0
0.5
1.0
1.5
2.0
2.5
3.0
t/T_ϕ
Numerical
Harmonic approximation